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import cv2
import numpy as np
from google.colab.patches import cv2_imshow # Import cv2_imshow

# Create a black image
img = np.zeros((500, 500, 3), dtype=np.uint8)

# Draw a white rectangle
cv2.rectangle(img, (100, 100), (400, 400), (255, 255, 255), -1)

# Draw circles at the corners
cv2.circle(img, (100, 100), 8, (0, 0, 255), -1)
cv2.circle(img, (400, 100), 8, (0, 255, 0), -1)
cv2.circle(img, (100, 400), 8, (255, 0, 0), -1)
cv2.circle(img, (400, 400), 8, (0, 255, 255), -1)

# Source points
src_pts = np.float32([
    [100, 100],
    [400, 100],
    [100, 400],
    [400, 400]
])

# Destination points
dst_pts = np.float32([
    [50, 50],
    [450, 100],
    [100, 450],
    [420, 400]
])

# Compute Homography Matrix
H, status = cv2.findHomography(src_pts, dst_pts)

# Apply Homography
output = cv2.warpPerspective(img, H, (500, 500))

# Display using cv2_imshow
cv2_imshow(img)
cv2_imshow(output)

# cv2.waitKey(0) and cv2.destroyAllWindows() are not needed with cv2_imshow in Colab
```

